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Project On

Code Refactoring and Bug Fixing

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About me

- **Hi I'm Shridhar Naragund [IN1240913],I'm Intern at Innomatics Research Lab , pursuing Data Science Course .**
- **Feel free to reach out! I'm always down to connect with you all. ☺**

  GitHub

Objective:

The objective of the debugging process was to identify and rectify issues in the existing codebase of a Note Taking Application developed using Python, Flask, and HTML.

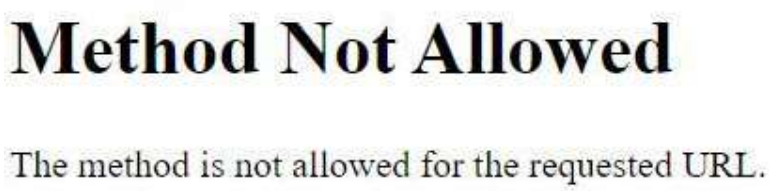
Throughout the debugging process, several issues were identified in the existing codebase. Below is a summary of each identified bug and the corresponding solutions:

Bugs Fixed:

Incorrect Request Method Handling in Flask:

Original Issue:

The Flask route was configured to only handle POST requests, neglecting GET requests. This limitation caused a method not allowed error when initially accessing the page.



Solution:

To address this, the allowed methods in the Flask route were updated to include both GET and POST methods using methods=["GET", "POST"]. This change enables the application to handle both types of requests.

Incorrect Form Method:

Original Issue:

The HTML form tag lacked the method attribute, resulting in the default method being "GET." This led to issues when submitting the form for adding notes.

```html
<body>
    <form action="">
        <input type="text" name="note" placeholder="Enter a note">
        <button>Add Note</button>
    </form>
```

Solution:

The form tag was updated to include method="post" to ensure that the form data is submitted using the POST method, which is essential for handling note additions.

```html
<form method="POST" action="/">
    <input type="text" name="note" placeholder="Enter a note">
    <button type="submit">Note Down</button>
</form>
```

Incorrect Retrieval of Note in Flask Route:

Original Issue:

The code for retrieving notes from the form used request.args.get("note"), which is suitable for GET requests but not for forms submitted via POST.

```
def index():
    note = request.args.get("note")
    notes.append(note)
    return render_template("home.html", notes=notes)
```

Solution:

The note retrieval code in the Flask route was corrected to use request.form.get("note"), ensuring the proper handling of note submissions through POST requests.

```
@app.route('/', methods=["GET", "POST"])
def index():
    if request.method == "POST":
        note = request.form.get("note")
        notes.append(note)
    return render_template("home.html", notes=notes)
```

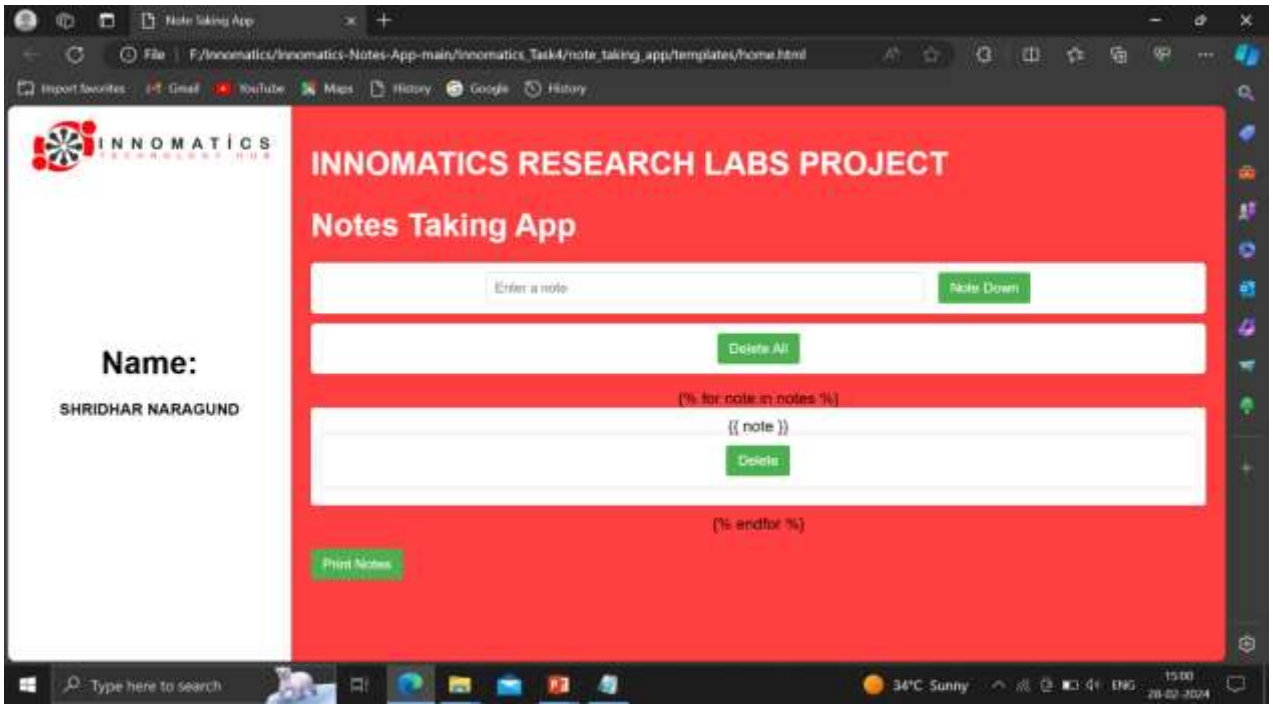
Flask Server Final:

```python
from flask import Flask, render_template, request, redirect

app = Flask(__name__)

notes = []

@app.route('/', methods=["GET", "POST"])
def index():
    if request.method == "POST":
        note = request.form.get("note")
        notes.append(note)
    return render_template("home.html", notes=notes)

@app.route('/delete/<int:index>', methods=["POST"])
def delete(index):
    if 0 <= index < len(notes):
        del notes[index]
    return redirect('/')

@app.route('/delete/all', methods=["POST"])
def delete_all():
    notes.clear()
    return redirect('/')

if __name__ == '__main__':
    app.run(debug=True)
```

```
            </ul>
            <button onclick="printNotes()">Print Notes</button>
        </div>
        <div class="right-sidebar">
       <img class="logo"
src="C:\\Users\\svani\\Downloads\\Innomatics_Task4\\note_taking_app\\Noteslogo
.jpg" alt="Right Sidebar Logo">
       <h2>Right Sidebar</h2>
       <p>Right sidebar content goes here.</p>
    </div>
    <script>
       function printAllNotes() {
           window.print();
       }
    </script>
</body>
</html>
```

Added Contact Details:

Original State: Contact details were not provided in the initial version.



Form Styling:

Enhancement:

Form styling was enhanced with a white background, border-radius, and a box shadow for a cleaner and more modern look. Adjustments were made to the form size and margin to improve its separation from the notes.

Print Functionality:

Enhancement:

A button to trigger the browser's print functionality was introduced using JavaScript. This addition allows users to easily print the notes.

```
<script>
    function printAllNotes() {
        window.print();
    }
</script>
```

Flask Routes:

Original State:

The Flask routes were limited to handling note additions only.

Enhancement:

Additional routes were introduced for deleting specific notes (/delete/<int:index>) and deleting all notes (/delete/all). These routes use the redirect function to navigate the user back to the main page after a deletion action.

Print Functionality Script:

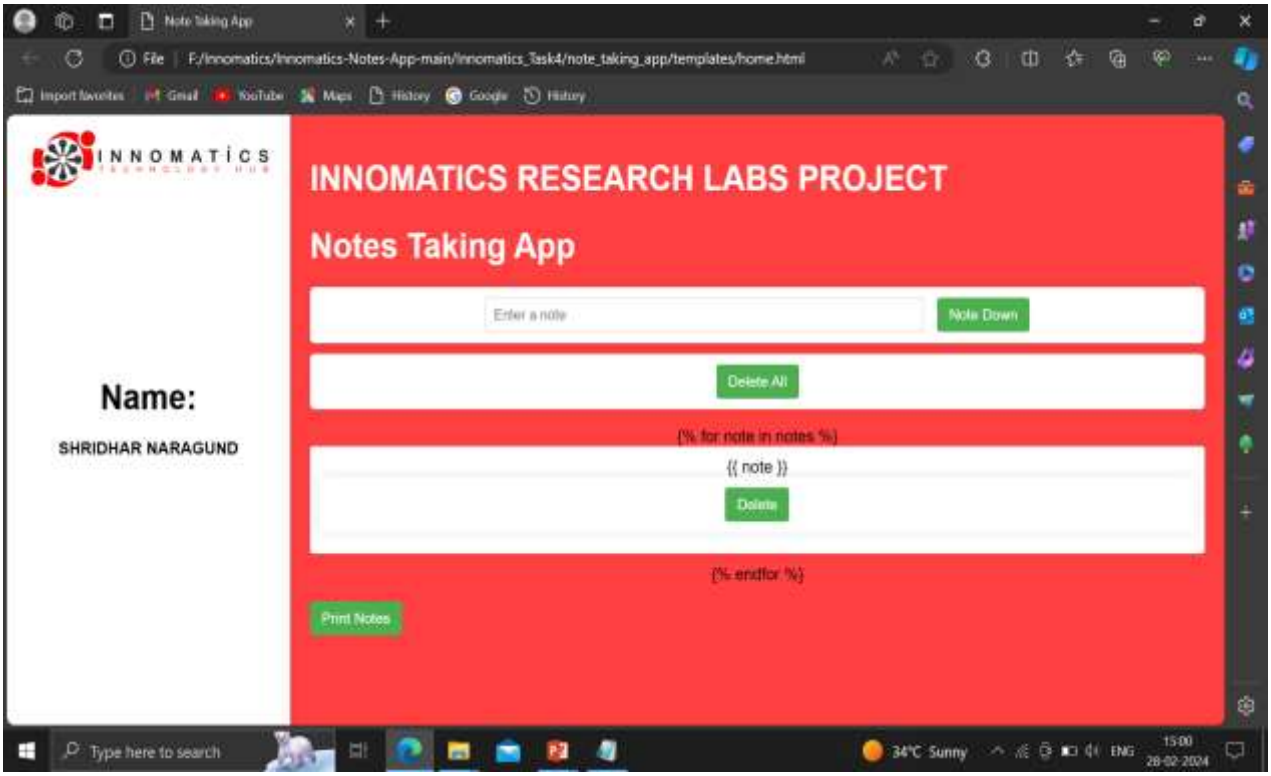
Original State:

The application did not have a script for print functionality.

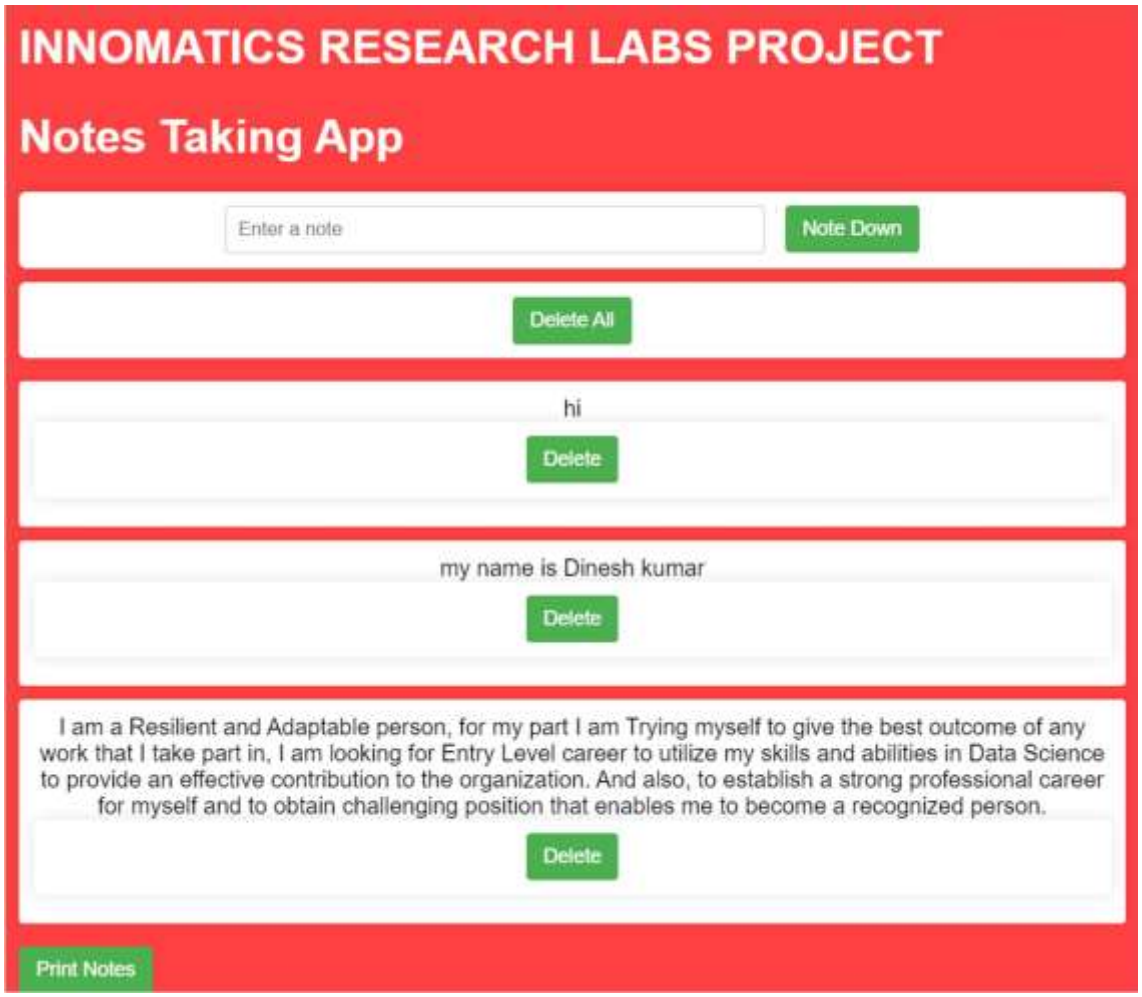
Enhancement:

A JavaScript function (printAllNotes()) was added to trigger the print functionality when the corresponding button is clicked. This script enhances the overall usability of the application.

UI:



Result:



Conclusion:

The debugging and refactoring process focused on addressing specific bugs related to form handling, request methods, and URL submissions. Simultaneously, enhancements were made to the user interface, styling, and functionality of the Note Taking Application. The result is a more polished and user-friendly application that not only fixes the identified issues but also adds new features and improves the overall user experience. The combination of improved HTML structure, CSS styling, and Flask routes contributes to a more robust and visually appealing
Note Taking App.

THANK YOU

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